



Society of Petroleum Engineers

SPE-227155-MS

Accelerating Multi-Dimensional Optimal Transport Full Waveform Inversion with Mini-Batch Optimization and Anisotropic Diffusion Filter

Hong Liang, Aramco Americas: Aramco Research Center-Houston; Young Seo Kim, Saudi Aramco, EXPEC Advanced Research Center; Vladimir Kazei and Weichang Li, Aramco Americas: Aramco Research Center-Houston

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227155-MS](https://doi.org/10.2118/227155-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Full waveform inversion (FWI) has become an industry standard tool for building high-resolution subsurface models. Conventional FWI using least-squares misfit function is known to suffer from cycle skipping and can be sensitive to noise. Recently, misfit functions based on the optimal transport (OT) have been proposed to replace the least-squares norm and show promising results in addressing the cycle skipping issue. Specifically, the multi-dimensional optimal transport FWI based on the Kantorovich-Rubinstein (KR) norm has demonstrated improved velocity update compared to least-squares FWI in various field data applications. However, solving for the KR norm in the multi-dimensional optimal transport is computationally quite expensive. In this paper, we present a workflow to accelerate multi-dimensional optimal transport FWI with mini-batch optimization and anisotropic diffusion filter.

Introduction

Seismic imaging plays a crucial role in exploring and characterizing complex subsurface structures. Full waveform inversion has emerged as an industry standard tool for obtaining high-resolution images of the subsurface models, especially in areas with complicated geological settings. Full waveform inversion formulated as a highly nonlinear data fitting procedure that iteratively updates the subsurface models by minimizing the misfit between recorded data and simulated data. The selection of misfit functions can significantly impact the convergence and efficiency of the inversion process. The least-squares norm is the most widely used misfit function for FWI. It measures the pointwise amplitude difference between recorded data and simulated data, and therefore is known to suffer from cycle skipping problem and also quite sensitive to noise in data.

There are many different approaches that have been developed to measure the data misfit based on other seismic attributes. The Wasserstein distance has recently been proposed as an alternative misfit function for FWI. It is originally defined through the optimal transport problem, measuring the minimum cost of transporting one distribution to another. In the context of FWI, the Wasserstein distance quantifies the discrepancy between observed and synthetic data. Results on both synthetic and real data indicate that

optimal transport FWI has the potential to mitigate the cycle skipping issue. Recent studies on optimal transport FWI (Engquist et al., 2016; Yang et al., 2018; Métivier, et al, 2016a; Métivier, et al, 2016b) have focused on two widely used metrics: the 2-Wasserstein distance and the 1-Wasserstein distance. Both the 2-Wasserstein and 1-Wasserstein distance are shown to have wider global minimum valley than the conventional least-squares norm. The 2-Wasserstein distance requires non-negativeness and unit mass for input data, necessitating data preprocessing or normalization for seismic traces (Yang et al., 2018; Qiu et al., 2017). Conversely, the 1-Wasserstein distance, often approximated by the Kantorovich-Rubinstein (KR) norm (Métivier et al., 2016a), does not impose the requirements of positivity and mass preservation on the input and therefore can be directly applied to seismic data without requiring data transformation.

In this paper, we focus on the optimal transport misfit function based on the KR norm. Computing the multi-dimensional KR norm involves solving a linear programming problem iteratively via simultaneous direction method of multipliers (SDMM), with each iteration of SDMM requiring solving a large matrix inverse problem. Therefore, multi-dimensional OT FWI based on the KR norm is computationally expensive. This paper introduces a workflow to reduce the computational cost of multi-dimensional optimal transport FWI based on the KR norm using mini-batch optimization and anisotropic diffusion filter. We will first review the theory behind the multi-dimensional optimal transport based on the KR norm. Then we present the proposed workflow and illustrate our method by using synthetic data examples.

Multi-dimensional optimal transport based on the KR norm

In this section, we review the numerical computation of the 1-Wasserstein distance based on the 2D KR norm. Following Métivier et al., (2016a) and Messud et al., (2021), the 2D Kantorovich-Rubinstein (KR) formulation of the 1-Wasserstein distance $\tilde{W}^1$ can be discretized as follow:

$$\left\{ \begin{array}{l} \tilde{W}^1(d_{cal}, d_{obs}) = \max_{\varphi} \sum_i \sum_j \varphi_{i,j} (d_{cal} - d_{obs}) \Delta t \Delta x_r \\ |\varphi_{i,j}| < \lambda \\ |\varphi_{i,j+1} - \varphi_{i,j}| \leq \frac{1}{\sigma_t} \Delta t \\ |\varphi_{i+1,j} - \varphi_{i,j}| \leq \frac{1}{\sigma_{il}} \Delta x_r \end{array} \right. \quad (1)$$

where d_{obs} and d_{cal} represent observed data and synthetic data, respectively. Δt and Δx_r are the discretization steps in the time and receiver coordinate, The discrete time and receiver coordinates are: $t_j = (j-1) \times \Delta t$, $j = 1, \dots, N_t$ and $x_i = (i-1) \times \Delta x_r$, $i = 1, \dots, N_x$, where N_t and N_x are the number of receivers and time samples in seismic traces. In equation 1, the absolute value of φ is bounded by the first constraint, the absolute value of the derivative of φ in the time direction is bounded by the second constraint, and the absolute value of the derivative of φ in the inline direction is bounded by the third constraint. The parameters σ_t and σ_{il} rescale the different physical quantities between time and inline positions, and account for balance between the derivative of φ in the time and inline directions. Detailed discussions about the proper choice of the constraint parameters ($\lambda, \sigma_t, \sigma_{il}$) can be found in the paper by Messud et al. (2021).

Solving equation 1 gives both the misfit value, representing the criterion to be maximized, and the source φ , the maximizer of the optimal transport problem. The numerical computation of the multi-dimensional KR norm is a linear programming problem, which can be solved iteratively using simultaneous direction method of multipliers (SDMM) (Combettes & Pesquet, 2011). Each iteration of SDMM requires the solution of a large linear system $\varphi^n = (I_N + A^T A)^{-1} \left[(y_1^n - z_1^n) + A^T (y_2^n - z_2^n) \right]$, which involves solving the inverse of the square matrix $Q = I_N + A^T A$, where Q is of size $N \times N$, and $N = N_t \cdot N_x$ in 2D. Solving this linear system requires a huge amount of memory associated with the matrix Q . For this reason, an alternative method is designed by Métivier et al., (2016b), which rewrite the linear system as a second-order finite difference discretization

of the Laplacian operator. The multigrid method (Adams, 1989) is then used to solve the elliptical partial differential equation. In summary, the convex non-smooth optimization problem in equation 1 associated with the 2D KR norm is iteratively solved by the SDMM algorithm, with each iteration of the SDMM requires the solution of an elliptic partial differential equation solved by the multigrid method. Therefore, the multi-dimensional OT FWI based on the KR norm is computationally expensive, especially for 3D applications.

Next, we first compare the conventional least-squares misfit function and the optimal transport misfit functions. Figure 1 illustrates the comparison of the least-squares misfit function (red), the 2-Wasserstein distance (blue), and the 1-Wasserstein distance based on the KR norm (black) for two shifted Ricker wavelets. The least-squares misfit function demonstrates multiple local minima. We can also observe that both the 2-Wasserstein and 1-Wasserstein distance have wider global minimum valley compared to the conventional least-squares misfit function, indicating the potential of the optimal transport misfit functions to reduce cycle skipping issues in FWI. We then compare the inversion results from the conventional least-squares FWI and the 2D optimal transport FWI using a synthetic example. Synthetic data are generated by finite difference modeling using the realistic 2D synthetic model shown in Figure 2a. For our experiment, a total of 96 sources with interval of 100 m and 501 receivers with interval of 20 m are placed at depth of 20 m. Synthetic data are generated using Ricker source wavelet of 6 Hz peak frequency. Figure 2b shows the initial velocity for FWI which is quite different from the exact model. Figure 3 compares the inversion results using the two misfit functions after 60 iterations (with the first 20 iteration focusing on early arrivals and the next 40 iteration focused on reflections). Figure 3a shows the least-squares FWI result, while Figure 3b shows the OT FWI result. Results from Figure 3 demonstrate the improved inversion result using the 2D KR misfit function compared to the conventional least-squares FWI.

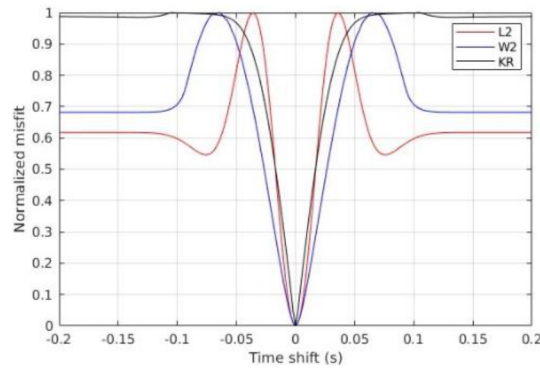


Figure 1—Comparison of least-squares norm (red), 2-Wasserstein distance (blue), and 1-Wasserstein distance based on the KR norm (black) for two shifted Ricker wavelets, where the horizontal axis is the shift value.

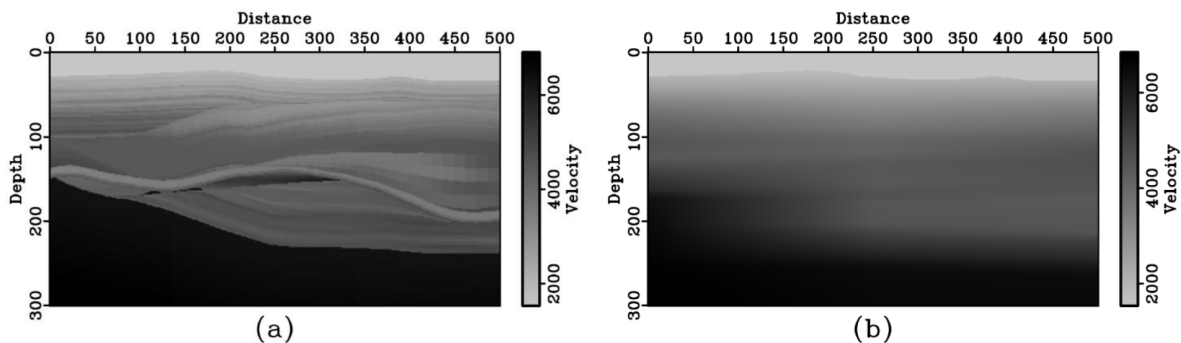


Figure 2—(a) Exact velocity model and (b) initial velocity model for FWI.

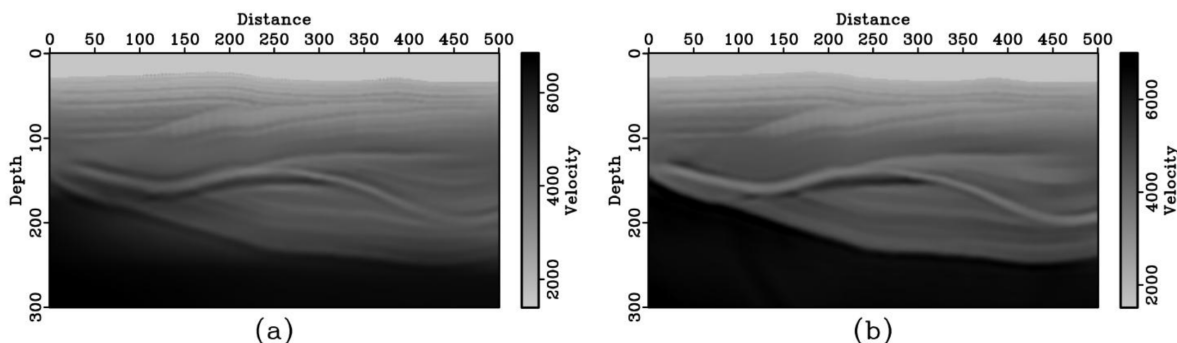


Figure 3—(a) Inversion result from least-squares FWI; (b) inversion result from OT FWI;

Mini-batch gradient descent optimization

The theory review in the previous section have indicated that the numerical computation of the 2D KR norm is computationally quite expensive. This section discusses the adoption of mini-batch gradient descent optimization (Yang et al., 2018; Kazei et al., 2023) to improve the efficiency of multi-dimensional OT FWI. For conventional FWI, the misfit function and the gradient are generated using all available data for each iteration. In the mini-batch optimization scheme, only subsets of the total data are used to compute the misfit function and construct the gradient at each iteration. Here, we use n to denote the size of the subset data chosen from the whole dataset (a total of N shots). For each iteration of the mini-batch optimization, the first shot is picked randomly and the rest of $n-1$ shots are then picked at equal intervals to ensure uniform coverage of the whole survey area. In FWI, each iteration involves multiple modeling for every input shot. Therefore, reducing the number of shots in the computation of the misfit function during each iteration directly decreases the computation time.

Next, we validate the above mini-batch OT FWI workflow on the previous synthetic dataset ($N=96$ in this case). We run two experiments with two different batch size: 8 shots ($n=8$) and 4 shots ($n=4$) for each iteration respectively. Figure 4 illustrates the inversion results from the mini-batch OT FWI inversion results with batch size of 8 shots (Figure 4a) and 4 shots (Figure 4b). It can be seen that both results show very similar structures as the inversion result using full shots (Figure 3b). The results demonstrate the effectiveness of the mini-batch optimization workflow. Compared to the conventional workflow using the all the shots in each iteration, results in Figure 4a only use $\frac{1}{12}$ of the total shots whereas Figure 4b only use $\frac{1}{24}$ of the total shots.

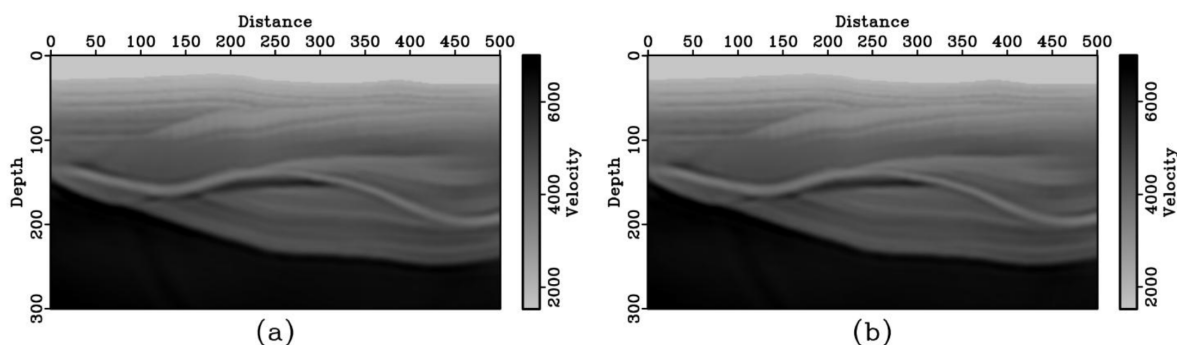


Figure 4—Mini-batch OT FWI inversion results with batch size of 8 shots (a) and 4 shots (b).

Anisotropic diffusion smoothing

Next we test the mini-batch OT FWI workflow on noisy data and apply non-linear anisotropic diffusion filtering (Weickert, 1998; Fehrenbach and Mirebeau, 2014; Métivier et al., 2024) for gradient smoothing. Originated from the field of image processing, anisotropic diffusion smoothing acts as a regularization technique for FWI when applied to smooth the FWI gradient. The elliptic Partial Differential Equation (PDE) form for the non-linear anisotropic diffusion is:

$$\frac{\partial u}{\partial t} - \nabla \cdot (\mathbf{D}(u) \nabla u) = 0, \quad (2)$$

where u at the initial time $t = 0$ is the input image, u at the final time $t = T$ is the smoothed image, $\mathbf{D}$ is the diffusion tensor. The diffusion tensor $\mathbf{D}$ can be defined in terms of the eigen-analysis of the structure tensor $\mathbf{S}$, where the structure tensor is constructed as:

$$\mathbf{S}(u) = K_\sigma * (\nabla u_\sigma \otimes \nabla u_\sigma), \text{ where } u_\sigma = K_\sigma * u, \quad (3)$$

where K_σ defines a Gaussian convolution kernel, the symbol $*$ denotes the convolution operator, and $\otimes$ defines the outer product operator. Let λ_i and $\mathbf{e}_i$ ($i=1,2$ in 2D) denote the eigenvalues and eigenvectors of $\mathbf{S}$.

$$\mathbf{S} = \sum_i \lambda_i \mathbf{e}_i \otimes \mathbf{e}_i \text{ and } \mathbf{D} = \sum_i \mu_i \mathbf{e}_i \otimes \mathbf{e}_i \quad (4)$$

Where the diffusion tensor $\mathbf{D}$ is constructed to share the same eigenvectors as the structure tensor $\mathbf{S}$, but with different eigenvalues μ_i that are designed to control the anisotropy of the diffusion process. Since the diffusion tensor $\mathbf{D}$ is derived from the image u , the resulting diffusion equation (Equation 2) is nonlinear. The matrix formulation of the structure tensor $\mathbf{D}$ enables directional smoothing, allowing the algorithm to preferentially smoothing along some directions. Following the methodology of Fehrenbach and Mirebeau (2014), the eigenvalues μ_i are chosen to enhance smoothing strongly along geological structures and gently across the structure, thereby preserving geologically meaningful structures while reducing noise and inversion artifacts.

The anisotropic diffusion smoothing relies on the information of the local structure. In the context of FWI, such structural information is not always readily available. In Yang et al. (2018), the local structure information is estimated from a synthetic fake image, which is obtained by high-pass filtering the true model and adding some random noises. In the application of Reflection full waveform inversion (RFWI), the inversion process involves alternatively updating a high wavenumber and a low wavenumber velocity model. Yao et al. (2019) obtained the local structure information from the high wavenumber velocity model, which is generated by the first step of RFWI. Building on these ideas, our approach constructs the stacked image using the starting model and all the shots. The local structure information is then estimated from the stacked image. The structural information derived from this image is computed once at the beginning of the inversion and held fixed throughout the process. Next we will show numerical examples to demonstrate the mini-batch workflow with the anisotropic diffusion filter.

Numerical examples

In this section, the noisy data is generated by adding noise to the previous synthetic data. Figure 5 compares one example of noise-free shot gather with the noisy shot gather. As described in the previous section, we use the stacked image to construct and structure tensor. to construct the anisotropic diffusion filter. The mini-batch OT FWI workflow is applied to the noisy data with the batch size of 4 shots. Prior to updating the velocity model, anisotropic diffusion smoothing is applied to the gradient for each iteration. To illustrate the comparison of different smoothing results, we apply three different smoothing method to the stacked image as shown in Figure 6. Figure 6a displays the raw stacked image from the noisy data using the initial model. Figures 6b and 6c show the results after isotropic Gaussian smoothing with window size of 200 m

and 100m, respectively. It is observed that isotropic smoothing with larger window size attenuates more noise but also but also blurs the images. Figure 6d illustrates the result after non-linear anisotropic diffusion smoothing, indicating enhanced noise reduction and preservation of structure integrity.

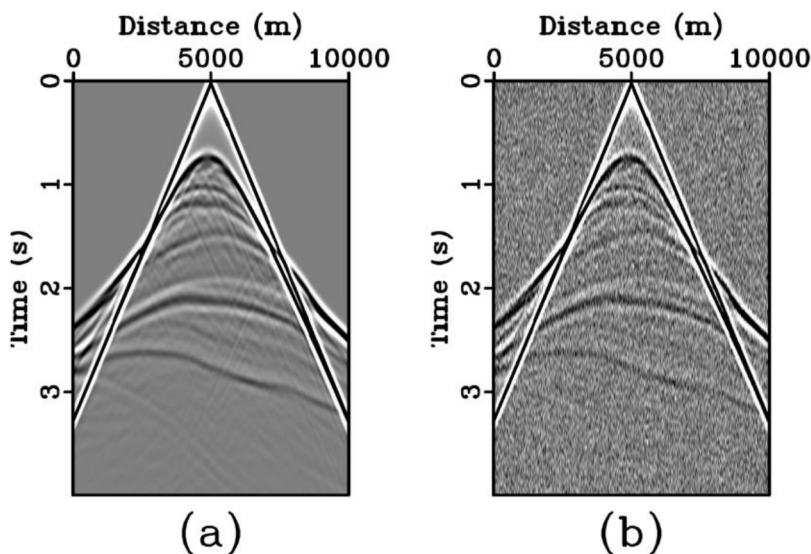


Figure 5—(a) Clean shot gather and (b) noisy shot gather.

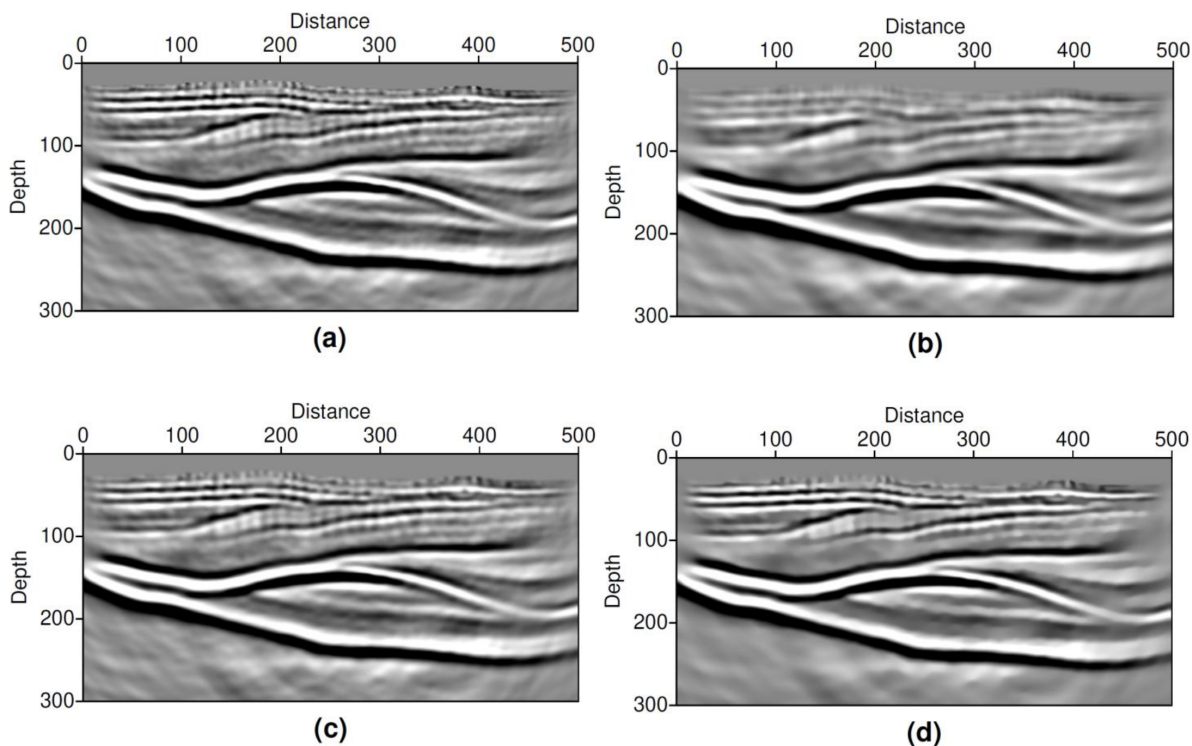


Figure 6—Comparison of results when different smoothing methods are applied to the stacked image: (a) raw image, (b) image after isotropic smoothing (window size of 200 m), (c) image after isotropic smoothing (window size of 100 m), and (d) image after non-linear anisotropic diffusion smoothing.

Figure 7 presents the final inversion results from noisy data using the proposed workflow. Figure 7a displays the inverted velocity model with isotropic smoothing (100 m window), while Figure 7b shows the result using non-linear anisotropic diffusion smoothing. For both results, the batch size is 4 shot per

iteration. From the comparison we can observe that the mini-batch OT FWI workflow with anisotropic diffusion smoothing gives a cleaner result with higher resolution.

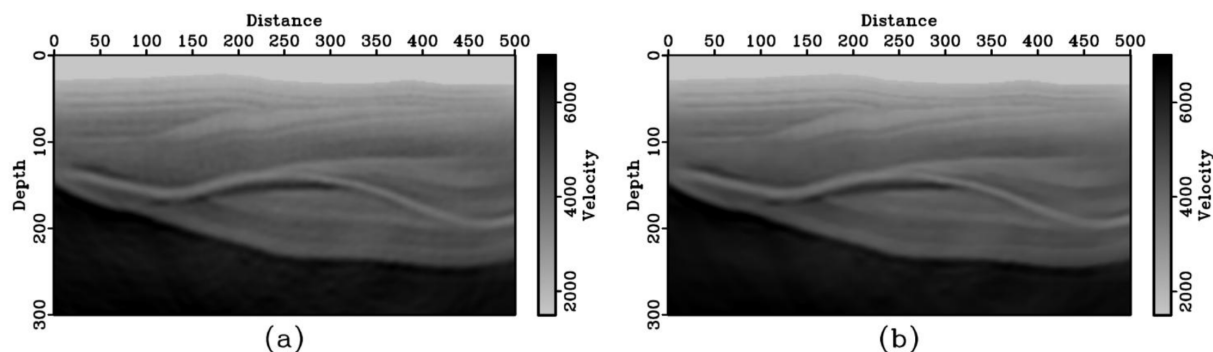


Figure 7—Inversion results from noisy data using mini-batch OT FWI: (a) with isotropic smoothing and (b) with non-linear anisotropic diffusion smoothing.

Conclusions

We have shown that the optimal transport FWI based on the KR norm is more robust against the cycle-skipping problem compared to the conventional least-squares FWI. However, computing the multi-dimensional KR norm itself is a convex non-smooth optimization problem that requires iterative solution, with each iteration involving a large matrix inverse problem. This makes the process computationally expensive and often impractical for large-scale 3D applications. To reduce the computational cost of the multi-dimensional OT FWI, we present a workflow using mini-batch gradient descent optimization with anisotropic diffusion filter. The mini-batch strategy reduces the number of shots involved in each iteration by using only a small, randomly selected subset of shots, while the anisotropic diffusion filter enhances the stability of the inversion by smoothing gradients along structural features. Numerical experiments validate the effectiveness of this approach. Results show the inverted model from the proposed workflow accurately recovers most structures when using only about 4% of the total shots for each iteration. The workflow significantly reduces the computational cost for the multi-dimensional optimal transport FWI based on the KR norm.

References

- Adams, J.C., 1989, MUDPACK: Multigrid portable FORTRAN software for the efficient solution of linear elliptic partial differential equations. *Applied Mathematics and Computation*, **34**(2):113–146.
- Combettes, P.L. & Pesquet, J.-C., 2011. Proximal splitting methods in signal processing, in *Fixed-Point Algorithms for Inverse Problems in Science and Engineering*, Vol. **49** of *Springer Optimization and Its Applications*, pp. 185–212, eds, Bauschke, H.H., Burachik, R.S., Combettes, P.L., Elser, V., Luke, D.R. & Wolkowicz, H., Springer.
- Engquist, B., B.D. Froese, and Y. Yang, 2016, Optimal transport for seismic full waveform inversion: *Communications in Mathematical Sciences*, **14**, 2309–2330.
- Fehrenbach, J., and J. Mirebeau, 2014, Sparse non-negative stencils for anisotropic diffusion: *Journal of Mathematical Imaging and Vision*, **49**, 123–147.
- Kazei, V., Liang, H. and AlDawood, A., 2023. Acquisition and near-surface impacts on VSP mini-batch FWI and RTM imaging in desert environment. *The Leading Edge*, **42**(3), pp.165–172.
- Métivier, L., R. Brossier, Q. Méridot, E. Oudet, J. Virieux, 2016a, Measuring the misfit between seismograms using an optimal transport distance: application to full waveform inversion: *Geophysical Journal International*, **205** (1), pp.345–377.
- Métivier, L., R. Brossier, Q. Méridot, E. Oudet, and J. Virieux, 2016b, An optimal transport approach for seismic tomography: *Application to 3D full waveform inversion: Inverse Problems*, **32**, 115008.
- Messud, J., R. Poncet, and G. Lambaré, 2021, Optimal transport in full-waveform inversion: analysis and practice of the multidimensional Kantorovich-Rubinstein norm: *Inverse Problems*, **37**.

- Métivier, L., R. Brossier, A. Hoffmann, J.-M. Mirebeau, G. Provenzano, A. Tarayoun, and P. Yong, 2024, Coherence-enhancing anisotropic diffusion filter for 3D high-resolution reconstruction of P-wave velocity and density using full-waveform inversion: Application to a North Sea ocean bottom cable data set, *Geophysics*, **89**, R33–R58.
- Qiu, L., J.R. Martínez, A. Valenciano, Y. Yang, B. Engquist, 2017, Full waveform inversion with an exponentially-encoded optimal transport norm: *SEG technical program expanded abstracts*, 1286–1290.
- Weickert, J., 1998, Anisotropic diffusion in image processing. Stuttgart Teubner.
- Yang, H., Jia, J., Wu, B. and Gao, J., 2018. Mini-batch optimized full waveform inversion with geological constrained gradient filtering: *Journal of Applied Geophysics*, **152**, pp.9–16.
- Yang, Y., B. Engquist, J. Sun, and B. Hamfeldt, 2018, Application of optimal transport and the quadratic Wasserstein metric to full-waveform inversion: *Geophysics*, **83**, R43–R52.
- Yao, G., N.V. Silva, D. Wu, 2019, Reflection-waveform inversion regularized with structure-oriented smoothing shaping. *Pure and Applied Geophysics*, 5315–5335.